System Requirements

P06: Anomalous Login Detection using ELK (Security Project)

<team member names & ids>

Student ID	Name
26100015	Muhammad Aaffan Khan Niazi
26100286	Mohammad Mustafa
25100022	SHEHROZ FARYAD
26100399	Mustafa Hussain

TABLE OF CONTENTS

1. Introduction

This project focuses on detecting anomalous login activities using the ELK stack (Elasticsearch, Logstash, Kibana) integrated with Wazuh. The system is designed to monitor authentication logs from multiple platforms such as Windows and Linux, process them, and analyze login behavior in real time to identify potential security threats.

Authentication logs are collected and forwarded through Logstash, where they are parsed, normalized, and enriched with relevant information such as timestamps and IP addresses. The processed data is then stored in Elasticsearch, enabling efficient searching and large-scale analysis. Kibana is used to visualize this data through dashboards, allowing users to easily monitor login activity and detect unusual patterns.

The system is capable of identifying various types of anomalies, including repeated failed login attempts, brute force attacks, unusual login times, and logins from unfamiliar locations or devices. When such activities are detected, alerts are generated to notify security personnel for further investigation and response.

The primary users of the system include Security Engineers, SOC Analysts, and System Administrators. These users interact with the system to configure detection rules, monitor alerts, investigate suspicious activities, and maintain system functionality. End users also benefit from improved account protection through continuous monitoring and timely alerts.

Overall, the system enhances organizational security by providing real-time visibility into login activities, reducing manual effort in log analysis, and enabling faster detection and response to potential threats.

2. System Actors

<List down the names of the system actors and give a 2-3 lines description of the role of each actor>

Actor Name	Description
Security Engineer	Configures detection rules, reviews anomalies, manages alerts.
SOC Analyst	Monitors dashboards, investigates anomalies, and responds to incidents.
System Administrator	Provides system-level logs, manages user accounts, and oversees infrastructure security.
End User	Generates login activity that the system monitors for anomalies.

3. Functional Requirements

<Write system requirements from users' (actors) perspective. Actor names have been highlighted in the sample requirements below. >

Requirements of Security Engineer	
Sr#	**Requirement**
1	I want to define anomaly detection thresholds so that suspicious logins can be flagged.
2	I want to manage detection rules to adapt to emerging attack vectors.
3	I want to view dashboards for anomaly trends to support investigations.
4	I want to tune detection rules based on false positives so that alert accuracy is improved.
5	I want to test detection rules on historical data so that I can validate their effectiveness before deployment.

Requirements of SOC Analyst	
Sr#	**Requirement**
1	I want to monitor real-time dashboards so that I can quickly identify suspicious login activities.
2	I want to investigate anomalies using Wazuh reports so that I can identify brute force or malicious login patterns.
3	I want to receive alerts for suspicious activities so that I can respond to incidents promptly.
4	I want to analyze login attempts across different systems so that I can correlate potential threats.
5	I want to escalate confirmed security incidents so that appropriate mitigation actions can be taken.
6	I want to add notes and update alert status so that investigation progress can be tracked efficiently.

Requirements of System Administrator	
Sr#	**Requirement**
1	I want to provide authentication logs to ensure comprehensive monitoring.
2	I want to manage accounts and permissions to reduce exposure to malicious logins.

3	I want to ensure system logs are continuously available so that no critical data is missed.
4	I want to maintain system uptime and reliability so that monitoring services remain uninterrupted.

Requirements of End User	
Sr#	**Requirement**
1	I want my login activity to be monitored to protect my account from unauthorized access.
2	I want to receive anomaly alerts so that I can take action to secure my account if suspicious activity occurs.
3	I want to be notified of failed login attempts so that I am aware of potential unauthorized access attempts.
4	I want to securely authenticate so that my account security is strengthened.

4. Non-functional Requirements / Quality Attributes

<Requirements must be testable>
<Security requirements fall in the category of “Non-functional requirements”; however, you need to list them separately in the section **Security Requirements** later in this document.>

Sr#	Requirements
1	The system should not utilize more than 1 GB of memory at any time during its execution.
2	The system should not fail more than 3 times every 24 hours; if it does, it should recover within 5 minutes.
3	The system should be able to process at least 10,000 log entries per second without performance degradation.
4	The alerting mechanism should deliver notifications within 30 seconds of anomaly detection.

5. Security Requirements

< Go through OWASP top 10 security risks in the following categories:

I. OWASP Top Ten: https://owasp.org/www-project-top-ten/
II. OWASP Mobile Top 10: https://owasp.org/www-project-mobile-top-10/
III. OWASP Machine Learning Security Top Ten: https://owasp.org/www-project-machine-learning-security-top-10/
IV. OWASP Top 10 API Security Risks: https://owasp.org/API-Security/editions/2023/en/0x11-t10/

(a) Select **security risks** that you think are primary threats for your system. While doing this, carefully consider the information/functionality that is most vulnerable from security perspective in the context of your project.
(b) For each security risk (identified above), identify **potential losses** (e.g., financial loss, total business loss, litigation etc.) if you do not take necessary measures to address the identified security risks.
(c) Identify the **controls** (e.g., input validation, audit logs, multi-factor authentication, user roles etc.) that should be implemented in your system to address the identified security risks.

Sr #	Security Risks	Potential Losses	Controls
1	Broken Access Control	Sensitive user login data is exposed.	Only security engineers will have update rights.
2	Input Manipulation Attack	Repeated attempts can cause the system to ignore the real threats.	Simulate the fake inputs to ensure the system's accuracy remains unaffected.
3	Data Poisoning Attack	Can cause the system's detection quality to drop over time.	Pre-process the data.
4	Using outdated versions of client-side and/or server-side components	Exploitation of known vulnerabilities in the older versions.	Obtain components from their official links
5	Hardcoding credentials	Credentials are exposed so hackers may be able to gain access	A security testing process would take place in order to ensure credentials are not exposed in such ways.

6. Security Engineer

< Each team must designate one member as the **Security Engineer**. While the entire team is responsible for implementation of the project's security features, the Security Engineer will take the lead in overseeing and ensuring the overall security of the project. **>**

Name of the Security Engineer	Muhammad Aaffan Khan Niazi

7. Use of Generative AI

<Mention here how generative AI was used in preparation of this artifact.>

8. Who Did What?

Name of the Team Member	Tasks done
Affan	Introduction, Actors
Mustafa Hussain	Functional requirements, review
Muhammad Mustafa	Security Requirements
Shehroz	Non Functional requirements

9. Review checklist

Before submission of this deliverable, the team must perform an internal review. Each team member will review one or more sections of the deliverable.

Section Title	Reviewer Name(s)
Introduction	Muhammad Aaffan Khan Niazi
Actors	Muhammad Aaffan Khan Niazi
Functional Requirements	Mustafa Hussain
Non-functional requirements	Shehroz Faryad
Security Requirements	Mohammad Mustafa
Use of Generative AI	